



Understanding Knee Replacement

Conventional, robotic-assisted & day-care (ERAS) surgery — everything you need to know before, during and after surgery.

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M.S. Orthopaedics · Joint Replacement,
Arthroscopy & Trauma

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BEFORE YOU BEGIN

What this guide covers

A complete walk-through — from recognising the symptoms, to choosing partial or total replacement, to the physiotherapy that carries you home.

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| 01 Do you need a knee replacement? | 05 Implants, materials & robotic systems |
| 02 Understanding your knee — basic anatomy | 06 Day-care joint replacement — the ERAS approach |
| 03 Types of knee replacement surgery | 07 Your journey — pre-op to post-op |
| 04 Robotic-assisted knee replacement — how it works | 08 Physiotherapy directions & frequently asked questions |



ABOUT YOUR SURGEON

Dr. Harshil Shah

Consulting Orthopaedic Surgeon specialising in joint replacement, arthroscopy and trauma. This guide reflects the conventional, robotic-assisted and day-care techniques used in his Ahmedabad practice — so you can walk into your consultation already informed.

M.S. ORTHOPAEDICS

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Every patient's anatomy, health status and recovery pace are different — use this guide to prepare better questions, not to replace a consultation.

1 GETTING STARTED

Do you need a knee replacement?

Knee replacement is most often recommended for osteoarthritis. It does not replace the whole joint; it resurfaces the worn ends of the bones — similar to how a dentist caps a worn tooth — while the surrounding ligaments and tendons are preserved.

TWO THINGS GENERALLY NEED TO BE TRUE

Significant cartilage loss — usually confirmed on a standing (weight-bearing) X-ray showing narrowing of the joint space.

Meaningful impact on life — persistent pain, difficulty with stairs or walking, or being unable to stay active despite trying non-surgical measures.

Try non-surgical options first — for many patients, anti-inflammatory medication, targeted strengthening exercises and joint injections provide real relief and surgery is not needed. Surgery becomes worth discussing once these measures stop being enough.

Partial vs. total — the knee has three compartments: inner (medial), outer (lateral) and the kneecap joint (patellofemoral). If arthritis is confined to just one compartment, a partial knee replacement may be possible, with a generally faster recovery.

9/10

patients have wear in more than one compartment and need a total knee replacement, where all three surfaces are resurfaced.



Symptoms and function matter as much as the X-ray.



Non-surgical measures are worth trying first.



One or three compartments changes what surgery looks like.

2 THE BASICS

Understanding your knee

The body's largest joint, formed where three bones meet — supported by ligaments and cushioned by cartilage. Surgeons think of it in three compartments.



- **Medial compartment**
Inner side of the joint — most commonly affected by wear.
- **Lateral compartment**
Outer side of the joint.
- **Patellofemoral compartment**
Where the kneecap glides against the femur.

Which compartments are affected determines whether a partial or total replacement is right for you.

F **Femur** — thigh bone, forms the upper part of the joint.

T **Tibia** — shin bone, bears most of your body weight.

P **Patella** — kneecap, glides in a groove on the femur.

C **Cartilage** — smooth coating that wears away in arthritis.

M **Menisci** — C-shaped shock absorbers, inner and outer.

L **Ligaments** — ACL/PCL inside, MCL/LCL on either side.

Synovium — a thin lining around the joint that produces fluid to lubricate and nourish it.

3 CHOOSING AN OPTION

Types of knee replacement surgery

Which type is right for you depends on how many compartments arthritis has affected, and the condition of your ligaments.

MOST COMMON

Total knee replacement (TKR) — all three compartments are resurfaced: the damaged cartilage and a thin layer of bone from the end of the femur, the top of the tibia and (usually) the underside of the kneecap, replaced with implant components. Recommended when arthritis affects more than one compartment.

FOR SELECT PATIENTS

Partial (unicompartmental) knee replacement — only the single damaged compartment, most often the medial (inner) one, is resurfaced. Healthy cartilage, bone and ligaments elsewhere — including the ACL and PCL — are left completely untouched.

WHEN IT'S A GOOD OPTION

Arthritis confirmed limited to one compartment only

ACL, and generally other main ligaments, intact and functioning

No significant knee deformity or major stiffness

No inflammatory arthritis affecting the whole joint

ADVANTAGES PATIENTS VALUE

Smaller incision, less soft-tissue disruption

Both cruciate ligaments preserved — a more "natural" feel

Typically faster early recovery than a total replacement

Usually a shorter stay, a good fit for day-care

Your surgeon will confirm suitability — ligament condition, alignment and the pattern of wear all need assessing first. If more than one compartment is found affected at surgery, your surgeon may need to proceed with a total knee replacement instead — this possibility is normally discussed with you in advance.

3 TYPES — CONTINUED

Bilateral & revision surgery

BOTH KNEES

Bilateral knee replacement

Both knees replaced, in one operation or as two separate surgeries — suitable only for select, medically fit patients, as recovery is more demanding when both are done together.

A REDO

Revision knee replacement

A redo procedure to replace a worn-out or loosened implant from a previous surgery; robotic assistance is used more selectively for this type of surgery.

4 ROBOTIC-ASSISTED KNEE REPLACEMENT

How it works

A surgical robot is one of several tools a surgeon can use — it does not perform the operation. The surgeon remains in direct control of every step; the robot assists with precision.

01

Registration & assessment

Optical trackers map your knee's exact anatomy in real time, and the surgeon checks how your ligaments behave through a range of motion.

02

Virtual planning

Software uses that data to plan implant position and size to a fraction of a millimetre, tailored to your own anatomy.

03

Guided execution

The surgeon uses robotically guided instruments to carry out the plan precisely, confirming fit and function with trial implants before placing the final ones.

5 MATERIALS & TECHNOLOGY

Implants, materials & robotic systems

Almost anyone having a first-time knee replacement can be a candidate for the robotic-assisted approach; it is used more selectively for revision surgery.

FEMORAL & TIBIAL

Metal components — femoral: a smooth metal shell, usually cobalt-chromium (sometimes titanium), shaped to replicate the natural curve of the thigh bone. Tibial: a metal tray, typically titanium or cobalt-chromium, that anchors into the shin bone.

BEARING & FIXATION

Plastic & cement — a high-density medical-grade polyethylene insert sits between the components, acting like the cartilage it replaces. Implants are secured with bone cement, or in select cases a cementless, porous-coated implant that allows bone to grow directly onto its surface.



All implant materials used are biocompatible and have a long track record of safe use in joint replacement surgery worldwide.

ROBOTIC SYSTEMS IN USE TODAY

MAKO

Stryker

CORI

Smith & Nephew

ROSA

Zimmer-Biomet

VELYS

DePuy-Synthes

Each works on the same broad principle — real-time tracking and virtual planning that guide the surgeon's instruments — with differences in hardware and workflow.

Surgeons using these systems require additional dedicated training and certification beyond their core joint-replacement training.

5 MATERIALS — CONTINUED

Conventional vs. robotic-assisted technique

MANUAL

Conventional technique

Surgeon uses manual jigs and instruments, guided by experience and intra-operative assessment

Bone-cut positioning planned and executed freehand, with proven accuracy in experienced hands

Ligament balance assessed by feel during surgery

Decades of long-term outcome data across all major implant systems

1-year satisfaction: high, in the 80–90% range

VS

PRECISION-GUIDED

Robotic-assisted technique

Surgeon uses the same clinical judgement, supported by a 3-D map and real-time tracking

Bone-cut planning and execution guided to sub-millimetre / sub-degree targets

Ligament tension measured and fine-tuned virtually before any cut is made

Growing evidence of improved alignment accuracy and, in some studies, faster early recovery

1-year satisfaction: similarly high, in the 80–90% range

A note on choosing between the two — outcomes after knee replacement are excellent overall, whether or not a robot is used — the technology is a tool, not a guarantee of a better result. Don't choose a surgeon or hospital on marketing claims that robotic surgery is automatically "better." Ask your surgeon directly which approach they recommend for you and why.

80–90%

one-year patient satisfaction, for either technique — this is the headline number worth remembering.

6 RECOVERY PATHWAY

Day-care joint replacement — the ERAS approach

Enhanced Recovery After Surgery (ERAS) is an internationally recognised pathway — first adopted for hip and knee replacement in Denmark and the UK — built around one idea: patients recover faster and more safely when kept active participants in their own recovery.

TRADITIONAL INPATIENT

Stay	Typically several days
Pain control	Leans more on opioid medication
Catheters/drains	More commonly used, kept in a period
Mobilisation	Usually begins the day after surgery

DAY-CARE / ERAS

Discharge	Same day, once checkpoints are met
Pain control	Opioid-sparing — opioids only as backup
Catheters/drains	Avoided, or removed at earliest safe point
Mobilisation	Within hours of surgery, under supervision

Day-care is not automatically right for everyone — you'll only be discharged once you clear defined checkpoints: pain controlled without IV medication, able to walk a safe distance and manage stairs, able to pass urine normally, and a responsible adult to support you at home. If any checkpoint isn't met, staying an extra night is a normal, safe part of the pathway.



Pain controlled without IV



Walk safely, manage stairs



Passing urine normally



Support at home available

7 THE FULL PICTURE

Your journey — pre-op to post-op

A structured, checkpoint-based path from your first consultation through to full recovery.

01 BEFORE SURGERY

Consultation

Clinical examination and standing X-rays confirm the diagnosis and extent of arthritis. Non-surgical options are discussed first; if surgery is right for you, conventional vs. robotic-assisted technique and day-care eligibility come next.

02 1-3 WEEKS BEFORE

Pre-operative optimisation

Blood tests, anaesthesia fitness check, and correction of issues such as anaemia if present. Pre-operative counselling and education, plus home planning — support at home, removing trip hazards, and any equipment advised.

03 SURGERY DAY

Day of surgery

Arrival, final checks and multimodal anaesthesia planning. Surgery itself — conventional or robotic-assisted, as planned — typically takes about one to two hours.

04 0-6 HOURS POST-OP

First hours after surgery

Pain control begins immediately using an opioid-sparing protocol. Early mobilisation — most patients are supported to stand and take first steps within about two hours, under physiotherapist supervision. Discharge checkpoints are assessed if you're on the day-care pathway.

Four steps down, three to go — discharge, early recovery and the months that follow continue on the next page.

7 YOUR JOURNEY — CONTINUED

Discharge & ongoing recovery

05 SAME DAY, OR SHORT STAY

Discharge

Clear written instructions on wound care, medication, physiotherapy and warning signs. A follow-up appointment and physiotherapy plan are scheduled before you leave.

06 WEEKS 1–6

Early recovery at home

Structured physiotherapy exercises continue daily. Most patients walk without assistance, manage stairs, and resume light daily activities within about six weeks.

07 3–12 MONTHS

Ongoing recovery

Strength, range of motion and confidence continue to improve gradually. Full recovery — including return to more demanding activity — typically takes six months to a year.



Most patients say it was worth it. Recovery pace varies — your surgeon and physiotherapist will tailor checkpoints to you.

RECOVERY AT A GLANCE

6 weeks

Walking unassisted,
stairs, light daily activities

3 months

Driving, most normal
routines resumed

6–12 months

Full recovery, return to
demanding activity

8 GETTING YOU MOVING

Physiotherapy directions

Recovery pace varies from patient to patient — always follow the specific exercise plan and timelines given by your surgeon and physiotherapist.

BEFORE SURGERY

Pre-habilitation

Straight-leg raises and quadriceps-tightening exercises, as guided by your physiotherapist, to strengthen the muscles supporting the knee.

DAYS 1–14

Immediately after surgery

Gentle range-of-motion exercises within pain limits; ankle pumps and knee bends through the day; supervised standing and walking; icing and elevation to control swelling.

WEEKS 2–6

Building strength & motion

Progressive knee-bending and straightening exercises to restore range of motion; supported stair practice and balance work; walking aids gradually reduced.

BEYOND 6 WEEKS

Returning to activity

Strengthening exercises for the thigh and hip muscles to support the new joint long-term; low-impact activity — walking, swimming, cycling — progressing towards your normal routine.

Managing swelling — use the ice sleeves provided regularly, and follow the Quiet Knee recovery protocol for pacing exercise and daily activity, combined with your prescribed pain-relief plan.

1

Control pain & swelling

2

Restore knee movement

3

Rebuild quadriceps strength

4

Return to walking & activity

9 FREQUENTLY ASKED QUESTIONS

The questions patients ask most

UNDERSTANDING THE SURGERY

Is this the same as replacing the whole joint?

No. Only the worn surfaces are resurfaced — comparable to capping a tooth. Surrounding ligaments and tendons are preserved.

Does the robot perform the surgery?

No. It's a guidance tool — your surgeon remains in full control of every step, including all bone cuts and implant placement.

Will I definitely need a total replacement?

Not necessarily. If arthritis is limited to one compartment, a partial replacement with faster recovery may be an option.

Is robotic surgery guaranteed to be better?

No single technique guarantees a better outcome — both have excellent success rates. Surgeon experience remains most important.

DAY-CARE & IMMEDIATE RECOVERY

Am I a candidate for day-care surgery?

Many patients are, but it depends on overall fitness, other medical conditions, and reliable support at home.

Will I be in severe pain without opioids?

The ERAS multimodal approach controls pain effectively while reducing side effects like nausea and grogginess.

Can I really walk within two hours?

Yes. Under our early-mobilisation protocol, most patients stand and take first steps within about two hours, physiotherapist by their side.

What if I don't clear the checkpoints?

You simply stay in hospital until you do — a routine, expected part of a safe day-care pathway, not a complication.

WOUND CARE

Will I need stitches removed?

No — we close with a stitchless technique using surgical skin glue. The dressing can be peeled off 15 days after surgery.

What kind of incision will I have?

At the front of the knee, usually about 4 to 6 inches, whether a conventional or robotic-assisted technique is used.

9 FREQUENTLY ASKED QUESTIONS — CONTINUED

Daily life, recovery & risks

DAILY LIFE & ACTIVITY

Can I kneel after a knee replacement?

Yes, it's perfectly safe once your incision has fully healed. A cushion under the knee usually helps at first.

How long do I wear compression stockings?

Typically the first two weeks while you're up and moving — they help control swelling. Your care team will confirm exactly how long.

When can I resume sexual activity?

Whenever you feel comfortable — there's no fixed waiting period. A reasonable question to raise with your surgeon.

Can I walk up and down stairs?

Yes — you'll be taught safe stair technique before you leave. Most patients manage stairs without difficulty from day one.

RECOVERY & SUPPORT

What is the long-term success rate?

Roughly 80–90% of patients report being satisfied with their result at one year, whichever technique was used.

What if I fall at home during recovery?

If you can't get up on your own, call for emergency help immediately. Contact your surgeon if you have pain bearing weight afterward.

How soon can I drive?

Most patients drive again within about three to six weeks, once off narcotic pain medication and able to react and brake confidently.

When is my follow-up appointment?

Usually between two and six weeks after surgery, depending on your recovery — scheduled for you before you leave hospital.

SURGERY DETAILS & RISKS

Both knees at once, or one at a time?

In most cases it's safer to have one knee replaced at a time. Both together is possible for carefully selected, healthy patients.

What are the general risks?

As with any major surgery: infection, blood clots, stiffness, nerve or vessel injury, and implant wear over time — uncommon, and the ERAS protocol helps minimise them.



Questions before surgery? Bring this guide to your consultation.

Every patient's anatomy, health status and recovery pace are different. Use the questions this guide raised to make the most of your time with your surgeon — we'll walk through it together.

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